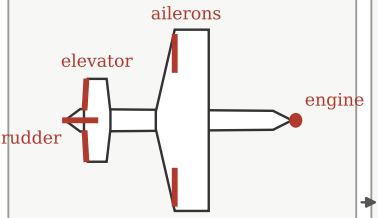
1 ALFA flight logs



Autopilot (Pixhawk) ⇄ Airframe + sensors

cmd →

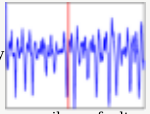
← meas

46 real flights, 10 Hz grid
servo outputs (rc-out) excluded

2 Physics-informed features

RES $r = \text{cmd} - \text{meas}$

roll, pitch, yaw, airspeed, altitude, cross-track, velocity



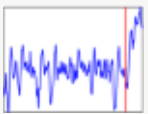
r_ϕ, aileron fault

CMD commands

$\phi^{\text{cmd}},\ \theta^{\text{cmd}},\ V^{\text{cmd}}$

EFF effectiveness

$g_a,\ \rho_a$: error vs. body rate



r_V, engine fault

3 s causal windows (0.5 s step), 5 statistics
⇒ 71 features per window

3 Detection, isolation, explanation

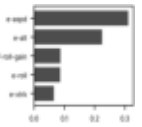
$\hat{P}(c \mid \mathbf{x}_w)$

Bagged trees (100)

Alarm: $1 - \hat{P}(\text{N}) \geq \tau$ for K windows

Isolation: Engine | Aileron
Rudder | Elevator

SHAP audit



compare with a priori fault signatures S_c

top-1 hit, mass M_c
⇒ consistent or shortcut

Leave-one-flight-out evaluation: detection rate, isolation, delay, false alarms per hour